SmartHome Gesture Control Application Project Part 2

Kai Chi

CSE535: Mobile Computing

Arizona State University

Feb 23, 2024

Intro

For part 2, we're asked to create a machine learning model to detect the gesture in a given video. We have 68 training videos and there are 17 different gestures.

Methodology

We start by coding up a feature extractor. It takes the middle frame of the video and resizes it to 200x200 in RGB. We get the class label from the file name of the video.

We use tensorflow to create a neural network. It takes an input of size 200x200x3. It does 2 rounds of convolution and 2 rounds of maxpooling. Then it flattens the array and feeds forward to a hidden, fully connected layer of 1000 neurons. The output layer size is 17x1. It uses the softmax function to produce the probabilities. The predicted class is the one with the highest probability. [1]

We specify a loss function in compile() and use the built-in train() function to train the model. The neural network learns the weights by using some form of gradient descent to minimize some loss function. [2] After 50 epochs, it has an accuracy of about 100% on the training set.

Result

We take this model and use it to do prediction on the test set. The accuracy is about 64%. If we have more training videos, we may be able to achieve a higher test accuracy score.

Reference

[1]

Built-in methods

https://www.tensorflow.org/guide/keras/training_with_built_in_methods

[2]

Adam optimization

https://www.tensorflow.org/api_docs/python/tf/keras/optimizers/Adam